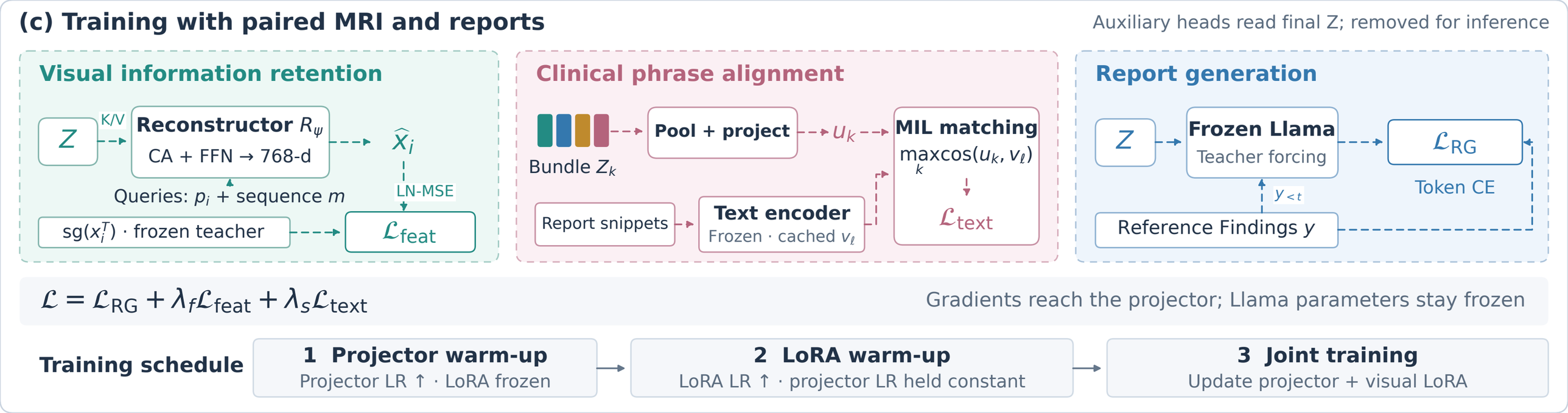

(c) Training with paired MRI and reports
Auxiliary heads read final Z; removed for inference
Visual information retention
Z
K/V
Reconstructor R_ψ
CA + FFN → 768-d
$\widehat{x}_i$
Queries: p_i + sequence m
LN-MSE
sg(x_i^T) · frozen teacher
$\mathcal{L}_{\text{feat}}$
Clinical phrase alignment
Bundle Z_k
Pool + project
u_k
MIL matching
$\max_k \cos(u_k, v_\ell)$
$\mathcal{L}_{\text{text}}$
Report snippets
Text encoder
Frozen · cached v_ℓ
Report generation
Z
Frozen Llama
Teacher forcing
$\mathcal{L}_{\text{RG}}$
Token CE
$y_{<t}$
Reference Findings y
$\mathcal{L} = \mathcal{L}_{\text{RG}} + \lambda_f \mathcal{L}_{\text{feat}} + \lambda_s \mathcal{L}_{\text{text}}$
Gradients reach the projector; Llama parameters stay frozen
Training schedule
1 Projector warm-up
Projector LR ↑ · LoRA frozen
2 LoRA warm-up
LoRA LR ↑ · projector LR held constant
3 Joint training
Update projector + visual LoRA